

Wavelet Augmented Phase Coherence Features for EEG-Based Imagined Speech Classification

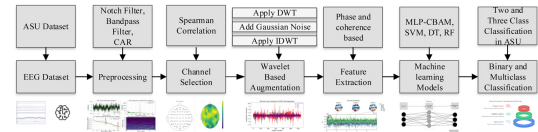
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Abstract—Brain-computer interfaces (BCIs) provide direct communication between the brain and external devices. Using electroencephalogram (EEG) sensors, BCIs are applied in assistive technologies and neuroprosthetics. Among various BCI paradigms, imagined speech-based BCI aims to decode internal speech representations from EEG signals, enabling silent communication. Decoding imagined speech is challenging due to the nonstationarity, intersubject variability of EEG signals and low signal-to-noise ratio. The proposed method uses a multilayer perceptron (MLP) integrated with a convolutional block attention module (CBAM) to enhance feature learning by refining spatial and channel-wise attention. To further improve performance, wavelet-based augmentation enhances data diversity. Phase and coherence-based functional connectivity features capture interchannel dependencies critical for imagined speech classification. The proposed wavelet-augmented phase coherence features with MLP-CBAM (WaveCoh-MLP-CBAM) framework is evaluated on an imagined speech dataset. The WaveCoh-MLP-CBAM shows superior classification accuracy, F1-score, and Cohen's kappa compared to conventional approaches. Results highlight the importance of augmentation, functional connectivity features, and attention in improving EEG-based imagined speech decoding.



Index Terms—Sensor signal processing, coherence, electroencephalogram (EEG), imagined speech, machine learning, phase, sensor signal processing.

I. INTRODUCTION

Brain-computer interfaces (BCIs) directly communicate between the human brain and external devices [1], [2]. BCIs depend on electroencephalogram (EEG) signals derived from sensors to decode user intentions without requiring physical movement. These systems are beneficial for individuals with motor impairments. Imagined speech-based BCIs offer a suitable method where users silently think about words, allowing for nonverbal communication without moving articulators. Decoding imagined speech from EEG signals is challenging due to the low signal-to-noise ratio (SNR), high intersubject and intertrial variability, and the weak neural activity associated with internal speech processes [3], [4]. EEG is the most widely used modality for BCI applications due to its noninvasiveness, high temporal resolution, and cost-effectiveness compared to other neuroimaging techniques such as functional magnetic resonance imaging (fMRI), magnetoencephalography (MEG) [5], and electrocorticography (ECoG) [6]. The fMRI provides superior spatial resolution, and MEG offers better signal localization. These fMRI and MEG methods require expensive and immobile equipment, limiting their real-world applicability. ECoG provides superior signal quality, is invasive and involves surgical implantation, making it unsuitable for widespread adoption. EEG captures real-time neural activity through scalp electrodes, making it an ideal choice for practical BCIs. EEG signals are highly sensitive to artifacts from muscle activity, eye blinks, and environmental noise, making robust preprocessing and feature extraction essential for accurate classification [7].

Several methods for imagined speech decoding have been explored in earlier studies [2], [4]. Qureshi et al. [8] used hybrid connectivity features, Kamble et al. [9] used smoothed pseudo wigner-ville distribution for imagined speech decoding. Reddy et al. [10] used dynamic mode decomposition and Lee et al. [1] used spectral features for imagined speech decoding. To improve EEG-based imagined speech classification, augmentation techniques are crucial in enhancing data and model robustness [11]. Traditional EEG classification models often struggle with limited training data, leading to overfitting and poor generalization. Augmentation methods such as wavelet-based transformations generate synthetic yet physiologically meaningful variations of EEG signals, enhancing model generalization and robustness. Applying wavelet transformations generates new synthetic samples while preserving the original frequency-domain characteristics [11]. Phase and coherence-based features have also proven to be highly effective in capturing the functional connectivity patterns within the brain during imagined speech. Coherence-based methods analyze the relationship between EEG channels, providing insights into functional connectivity. Imagined speech involves distributed cortical activation, phase analysis, and coherence-based connectivity, enhancing classification performance.

The proposed WaveCoh-MLP-CBAM framework introduces a novel combination of biologically inspired augmentation, functional connectivity features, and attention-based learning for imagined speech classification. Wavelet-based augmentation enhances data diversity by injecting noise into detail coefficients, maintaining spectral integrity while simulating realistic variability. Phase difference and spectral coherence features capture functional connectivity between EEG channels, reflecting neural synchrony. A multilayer perceptron with convolutional block attention module (MLP-CBAM) is employed to refine feature representation. The MLP captures nonlinear EEG

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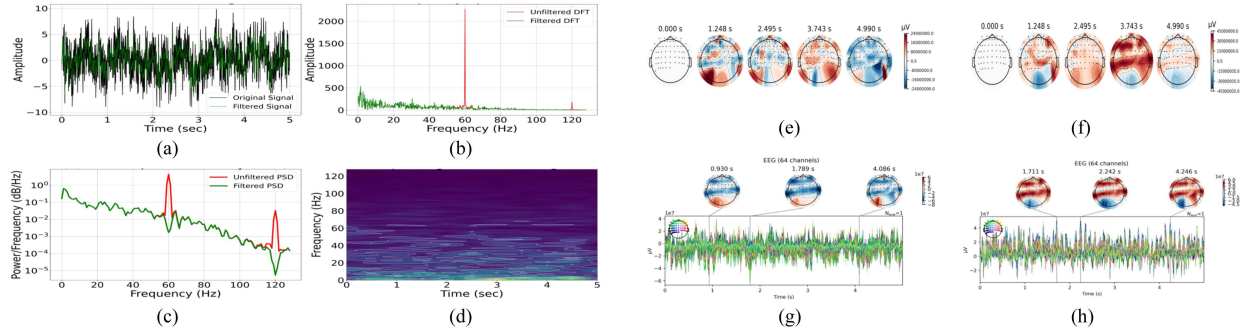


Fig. 1. Signal processing and evoked potential analysis for imagined speech classification. (a) Time-domain signal, (b) frequency spectrum using discrete fourier transform (DFT), (c) power spectral density, and (d) spectrogram illustrate preprocessing. (e)–(h) Show evoked potential responses distinguishing cooperative and independent words where (e) and (g) correspond to “cooperate” and (f) and (h) to “independent.”

dynamics, while CBAM highlights spatial and channel-level relevance. This approach enables the model to focus on the most informative features, improving classification performance.

II. MATERIAL AND METHODS

A. Dataset Used

The Human-Oriented Robotics and Controls (HORC) Lab at Arizona State University recorded this dataset [12]. EEG signals were collected using a BrainProducts ActiChamp amplifier with 64 electrodes, placed following the 10–20 system. The data were recorded at 1000 Hz and later down-sampled to 256 Hz. The dataset includes four types of imagined speech: long words (“independent,” “cooperate”), short words (“in,” “out,” “up”), vowels (“a,” “i,” “u”), and short–long words (“in,” “cooperate”). It was recorded from 15 healthy participants (11 males and four females, aged 22–32). Most completed 100 trials per word, while some had 80 trials due to protocol limits. To reduce noise, electrodes 1, 10, 33, and 64 were removed due to electrooculography (EOG) artifacts. Participants imagined speaking the words without moving, following beep cues for timing. The dataset contains 1200 long-word trials, 1120 short–long word trials, 1800 short-word trials, and 2400 vowel trials. This dataset is publicly available. It is useful for imagined speech decoding, BCIs, and EEG research.

B. Preprocessing and Channel Selection

The EEG preprocessing pipeline includes filtering, normalization, and channel selection to improve signal quality. A band-pass but-terworth filter (4–70 Hz, order 4) removes unwanted frequencies, followed by a Notch filter (60 Hz, 120 Hz, quality factor 30) to eliminate powerline interference [2], [9], [13]. Z-score normalization standardizes data across trials. Spearman correlation-based channel selection identifies the most relevant channels, and channel removal discards noisy or less informative ones. These steps reduce artifacts and enhance features for classification. Fig. 1 shows EEG preprocessing and feature extraction for imagined speech classification. Fig. 1(a)–(d) shows signal processing and Fig. 1(e)–(h) displays EEG responses. The topographic maps Fig. 1(e) and (f) highlights spatial EEG activity, Fig. 1(g) and (h) shows temporal variations. These patterns help distinguish imagined words using spatiotemporal features.

C. EEG Signal and Phase Based Features

This study use wavelet-based augmentation, phase-based, and coherence-based features to enhance the discriminability of EEG

signals. These methods help capture temporal, spectral, and phase-related characteristics critical for imagined speech classification. The key steps involved in the proposed methodology are detailed as follows.

- 1) *Wavelet Augmentation*: EEG signals are decomposed using the discrete wavelet transform (DWT)

$$x(t) = \sum_{j=0}^J c_j \phi_j(t) + \sum_{j=0}^J \sum_k d_{j,k} \psi_{j,k}(t) \quad (1)$$

where $x(t)$ is the EEG signal at time t , J is the decomposition level, $\phi_j(t)$ represents scaling functions with coefficients c_j , and $\psi_{j,k}(t)$ are wavelet functions with detail coefficients $d_{j,k}$ at scale j and position k . Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ is added to the detail coefficients

$$\tilde{d}_{j,k} = d_{j,k} + \epsilon \quad (2)$$

where $\tilde{d}_{j,k}$ is the noise-perturbed coefficient, and σ^2 is the noise variance. The augmented signal $\tilde{x}(t)$ is reconstructed via inverse DWT

$$\tilde{x}(t) = \sum_{j=0}^J c_j \phi_j(t) + \sum_{j=0}^J \sum_k \tilde{d}_{j,k} \psi_{j,k}(t). \quad (3)$$

- 2) *Phase-Based Features*: The Hilbert transform extracts the analytic signal

$$H(x(t)) = x(t) + j\mathcal{H}\{x(t)\} \quad (4)$$

where $H(x(t))$ is the analytic signal, $x(t)$ denote the EEG signal at time t , $\mathcal{H}\{x(t)\}$ is the Hilbert transform, and j is the imaginary unit. The instantaneous phase is

$$\theta(t) = \tan^{-1} \left(\frac{\mathcal{H}\{x(t)\}}{x(t)} \right) \quad (5)$$

where $\theta(t)$ is the phase at time t . The phase difference between EEG channels i and j is

$$\Delta\theta_{i,j}(t) = \theta_i(t) - \theta_j(t) \quad (6)$$

where $\Delta\theta_{i,j}(t)$ represents the phase synchronization between channels. Extracted features include the mean and variance of phase difference

$$\mu_\theta = \frac{1}{N} \sum_{t=1}^N \Delta\theta_{i,j}(t), \quad \sigma_\theta^2 = \frac{1}{N} \sum_{t=1}^N (\Delta\theta_{i,j}(t) - \mu_\theta)^2 \quad (7)$$

where μ_θ is the mean phase difference, σ_θ^2 is its variance, and N is the number of time points.

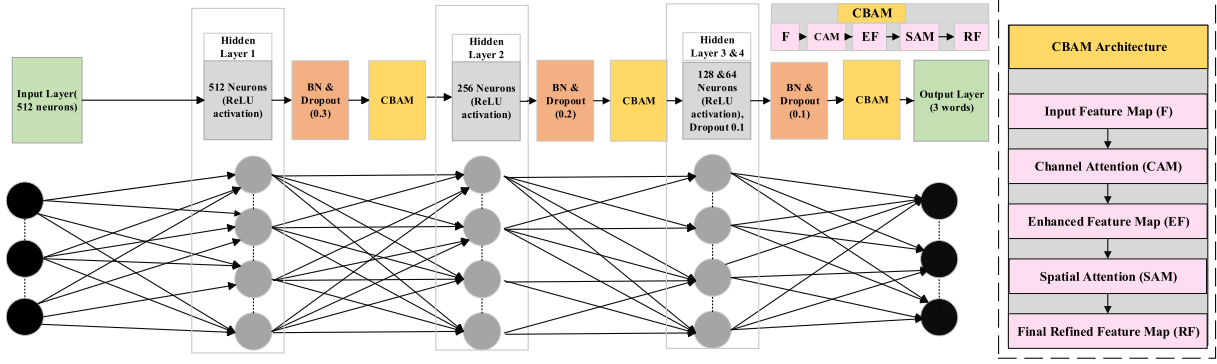


Fig. 2. Proposed MLP-CBAM model integrating MLP with CBAM for improved feature extraction and classification.

- 3) *Coherence-Based Features*: Coherence between EEG signals $x_i(t)$ and $x_j(t)$ at frequency f is computed as

$$C_{i,j}(f) = \frac{|S_{i,j}(f)|^2}{S_{i,i}(f)S_{j,j}(f)} \quad (8)$$

where $C_{i,j}(f)$ is the coherence, $S_{i,j}(f)$ is the cross-power spectral density between signals x_i and x_j , and $S_{i,i}(f)$, $S_{j,j}(f)$ are their respective auto-power spectral densities. Extracted features include

$$C_{\text{mean}} = \frac{1}{F} \sum_{f=1}^F C_{i,j}(f), \quad C_{\text{max}} = \max_f C_{i,j}(f) \quad (9)$$

where C_{mean} is the mean coherence, C_{max} is the maximum coherence, and F is the total number of frequency bins.

The model is trained and evaluated using a subject-dependent protocol, where each sample corresponds to a complete EEG trial. No trial is shared across the training, validation, or test sets, ensuring that all partitions remain temporally and contextually independent. A stratified split is applied to allocate 70% of the data to training, 15% to validation, and 15% to testing. This approach is aligned with recommendations from Li et al. [14] and Kamrud et al. [15].

D. Architecture of the Proposed Model

The model integrates MLP with CBAM to enhance feature learning as in Fig. 2. The input layer of the MLP-CBAM model has 512 neurons. It receives a one-dimensional vector formed by flattening the extracted functional connectivity features. These features are based on phase difference and coherence between selected EEG channels. The input layer transforms these features into a dense representation, which is then processed by attention modules in later layers. The MLP consists of four hidden layers with 512, 256, 128, and 64 neurons, each using ReLU activation for nonlinearity. Batch Normalization (BN) stabilizes training, and Dropout (0.3, 0.2, 0.1, 0.1) stops overfitting. CBAM includes convolutional neural network (CNN) based attention mechanisms applied at the 1st, 2nd, and 4th hidden layers to refine feature representations. The attention mechanism includes channel and spatial attention modules to highlight important features. The model is trained using the Adam optimizer (learning rate: 0.0005) and sparse categorical cross-entropy loss for 150 epochs with a batch size 32. The Softmax output layer does the classification task. The architecture enhances classification accuracy by integrating MLP for dense feature learning and CBAM for spatial and channel attention using CNN-based operations.

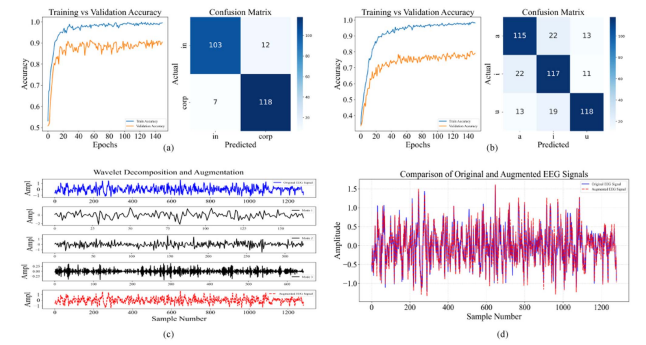


Fig. 3. (a) Training versus validation accuracy and confusion matrix for binary classification of "in" and "corporate"(corp). (b) Accuracy and confusion matrix for multiclass classification of imagined phonemes ("a", "i", "u"). (c) Wavelet decomposition showing different frequency modes for EEG augmentation. (d) Comparison of original and augmented EEG signals, highlighting augmentation effects.

III. RESULTS

The proposed method attained the highest performance across all evaluation metrics, significantly outperforming existing approaches as shown in Table 1. It obtained an accuracy of 85% and a Kappa score of 0.70 for long words, compared to 0.74 (VMB+EBT), 0.71 (ISFCG), and 0.66 (ESSV). Improvements are observed for short words, vowels, and short-long words, where the proposed method consistently outperformed other techniques. Fig. 3(a) and (b) shows the accuracy versus validation curve and confusion matrix for short-long words and vowels. Fig. 3(c) illustrates EEG augmentation, and Fig. 3(d) compares the original EEG with the segmented signal. The superior performance is attributed to three key factors. First, wavelet augmentation improved the EEG signal variability, improving generalization. Second, Phase and Coherence-based features effectively captured temporal synchronization and frequency-domain relationships, crucial for distinguishing imagined speech patterns. Third, the MLP-CBAM model, with BN and Dropout, improved feature learning by extracting complex patterns while CBAM enhanced spatial and channel-wise attention, preventing overfitting, and ensuring stability. Spearman correlation-based channel selection helped remove irrelevant information, further boosting classification accuracy. Combining these strategies enabled the proposed method to extract discriminative EEG features, leading to higher accuracy and robustness.

The results in Fig. 4(a) indicate that MLP-CBAM achieved the best performance for imagined speech EEG classification, with the highest

Table 1. Performance Comparison Across Techniques

Technique Used	Long Words			Short Words			Vowels			Short Long Words		
	Acc	F1-Score	Kappa	Acc	F1-Score	Kappa	Acc	F1-Score	Kappa	Acc	F1-Score	Kappa
ESSV [16]	0.66	0.66	0.32	0.42	0.42	0.14	0.44	0.44	0.16	0.62	0.62	0.23
Tangent + RVM [12]	0.66	-	-	0.51	-	-	0.49	-	-	0.80	-	-
ISFCG [2]	0.71	0.71	0.42	0.44	0.44	0.16	0.46	0.46	0.20	0.70	0.70	0.40
VMD + EBT [4]	0.74	-	-	0.72	-	-	0.66	-	-	0.74	-	-
Proposed Method	0.85	0.85	0.70	0.65	0.65	0.47	0.78	0.78	0.67	0.90	0.90	0.80

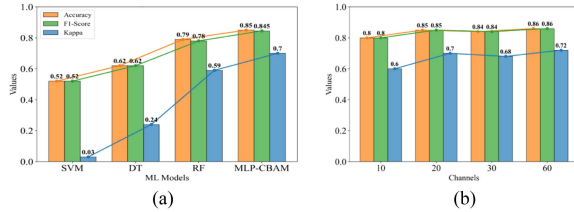


Fig. 4. (a) MLP-CBAM outperforms SVM, DT, and RF in imagined speech classification. (b) Fewer channels achieve high accuracy, while more channels offer slight improvement with higher cost.

accuracy (0.85), F1-score (0.845), and Kappa (0.70). To further confirm the consistency of the results, k -fold cross-validation with $k = 3, 5$, and 7 was performed. All evaluation metrics, including accuracy, F1-score, and Cohen's kappa, are computed on the held-out test set. The superior performance is due to CBAM's attention mechanism, which enhances spatial and channel-wise feature selection, allowing the model to focus on the most relevant EEG patterns. SVM, DT, and RF depend on handcrafted features, MLP-CBAM effectively captures complex neural representations.

The results in Fig. 4(b) also show that even with fewer electrodes (10 or 20), high accuracy (0.80–0.85) was achieved. It suggests the Spearman correlation effectively selects the most relevant EEG channels for imagined speech classification. Using fewer electrodes reduces computational complexity and training time while preserving key discriminative features. Increasing the number of channels (e.g., 60) slightly improves performance (0.86 accuracy) by capturing more spatial information at the cost of higher processing requirements.

IV. CONCLUSION

WaveCoh-MLP-CBAM combined channel selection, wavelet augmentation, phase and coherence features, and MLP-CBAM to improve accuracy. Spearman correlation-based channel selection removed irrelevant channels, reducing noise. Wavelet augmentation enhanced data diversity and model generalization. Phase and coherence features captured brain connectivity patterns important for imagined speech. MLP-CBAM enhanced feature learning by combining deep learning with attention mechanisms. The results showed that the proposed method outperformed existing approaches in accuracy, F1-score, and Cohen's kappa. Combining feature engineering, augmentation, and attention-based learning made the model more robust. Future work will focus on hybrid deep learning models, combining CNNs and transformers for better feature extraction. Variational autoencoders (VAEs) will be explored for EEG data augmentation. Multimodal approaches integrating fNIRS or MEG may enhance classification accuracy. Real-time implementation will be prioritized to improve the

practicality of imagined speech-based BCIs. The study uses a subject-dependent setup that effectively captures individual EEG patterns. While cross-subject performance is not evaluated, it will be explored in future work. The method includes offline steps like wavelet-based augmentation and feature extraction, which may require further optimization for real-time use.

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